Data Engineering

Nights of 2026-09-28 to 2026-09-30 · Demo Client

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Engagement	DEMO-0012
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Executive summary

Demonstration report. Built from synthetic data generated by ProQuant for illustration: a café-restaurant chain that does not exist, whose systems deliver three nights of files.

ProQuant Analytics built a data platform for Demo Client: every source is collected, checked against a written contract, kept as received, brought to standard tables, and published as tables that dashboards and reports can trust. It ran for 3 nights (2026-09-28 to 2026-09-30) before this report, including a night on which one system sent a broken file and another changed its format.

Overall view: The platform is ready to hand over, with actions

10 of 15 checks passed, 5 with a warning, 0 failed. The thresholds were agreed before the platform was built.

SOURCES	ROWS IN THE PLATFORM	NIGHTS RUN
6	40,381	3
1 read directly	6 standard tables	last: warnings

SET ASIDE LAST NIGHT	TOTALS AGREE	A NIGHT TAKES
1	yes	1.0 s
never half-loaded	staging to every published table	first load 161 s

Findings

- **Sources** 6 sources are collected: 1 read directly from their system, 5 through files dropped in a folder.
- **First load** The first night loaded 548 batches, 40,074 rows, in 161 seconds; later nights took 1.0 seconds at most.
- **Set aside** pos_B1_2026-09-30.csv (pos) was quarantined: required column missing: Net Sales. Nothing of it reached the published tables.
- **Format change** pos_B3_2026-09-30.csv (pos): new column(s): Tip; stored in the raw layer
- **Duplicate** pos_B1_resent_copy.csv was skipped: same content as pos_B1_2026-09-20.csv.
- **Correctness** Totals agree from staging to every published table to the cent.
- **Repeatable** Running again with nothing new loaded nothing and changed nothing.
- **Portable** The same models were built and tested on a second database.
- **Package** The platform package installs and passes its self-test on a clean folder.

Actions

Ref	Priority	Action	Owner	Due
A-01	Medium	Read the last run's quarantine notes and fix the sources at their origin.	Data owner	2026-11-14
A-02	Medium	Fix the quarantined files at source and drop them again; they load on the next night.	Data owner	2026-11-14
A-03	Medium	Find why the site or table is behind: a quarantined or missing file.	Data owner	2026-11-14
A-04	Medium	Ask each system's provider for an API or a read-only database user; replace manual drops one by one.	Data owner	2026-11-14
A-05	Medium	Agree the new format with the provider and add the new columns to the contract and the mapping.	Data owner	2026-11-14

Source: tables/actions.csv

Methodology

This report tells what was built and how it behaved. The design, the rules of each layer and every parameter are in the methodology document; daily operation is in the runbook. One thing helps when reading on: a batch is either loaded whole or set aside whole, so a published table never holds half a file.

Results

1. The sources

Each system, how the platform reaches it, and how it is loaded: new files once each, full exports replaced when they change, or new rows since the last watermark.

Source	Type	Collected	Load mode	Batches	Latest	Rows	Columns	Contract
pos	file_drop	drop folder	new_files	543	pos_B3_2026-09-28.csv	90	10	yes
menu	file_drop	drop folder	snapshot	1	menu_items.xlsx	9	5	yes
branches	file_drop	drop folder	snapshot	1	branches.csv	3	6	yes
ledger	file_drop	drop folder	snapshot	1	gl_export.csv	78	4	yes
ads	file_drop	drop folder	snapshot	1	ads_export.csv	549	7	no
hr	database	connected	snapshot	1	database read	18	5	no

Source: tables/source_inventory.csv

2. What was built

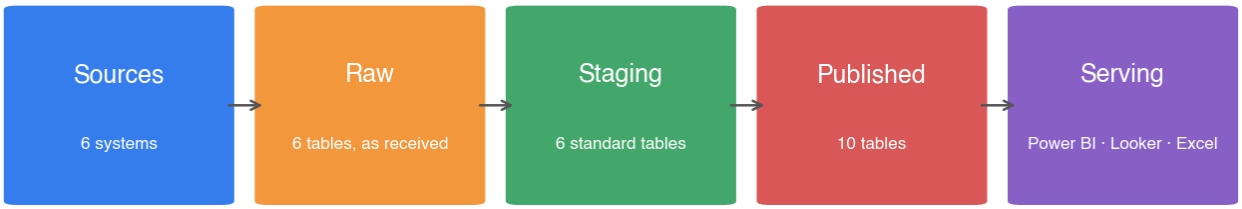


Figure 1. The layers of the platform. Source: tables/layers.csv

Table	Layer	Rows	Columns
_drift	control	1	5
_loads	control	572	10
_runs	control	4	12
_tests	control	108	5
dim_customer	published	899	5
dim_date	published	183	6
dim_item	published	9	6
dim_place	published	3	5
fact_daily_sales	published	548	9
fact_expenses_monthly	published	78	4
fact_marketing_daily	published	549	8
fact_sales	published	39,723	13
fact_staff_monthly	published	18	5
mart_kpi_monthly	published	18	10
raw_ads	raw	549	12
raw_branches	raw	3	11
raw_hr	raw	18	10
raw_ledger	raw	78	9
raw_menu	raw	19	10
raw_pos	raw	40,576	16
stg_expenses	staging	78	9
stg_items	staging	10	8
stg_marketing	staging	549	12
stg_places	staging	3	9
stg_staff	staging	18	11
stg_transactions	staging	39,723	21

Source: tables/layers.csv

3. Three nights

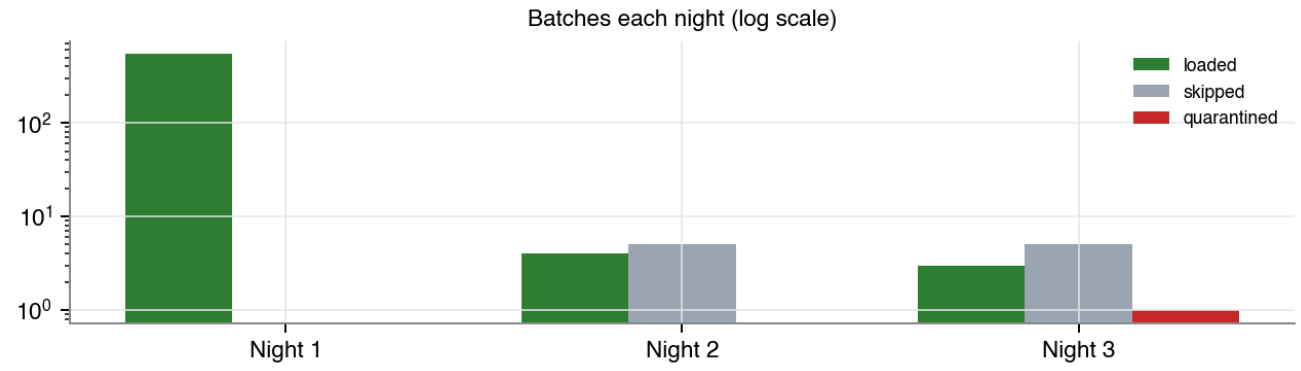


Figure 2. Batches loaded, skipped and set aside each night. Source: tables/loads.csv

Run	Night of	Status	Seconds	Loaded	Skipped	Set aside	Rows	Tables built	Failed	Warnings
night1	2026-09-28	ok	160.89	548	0	0	40,074	9	0	0
night2	2026-09-29	ok	1.04	4	5	0	241	9	0	0
night3	2026-09-30	warnings	0.88	3	5	1	123	9	0	1
rerun	2026-09-30	warnings	0.65	0	5	1	0	9	0	1

Source: tables/runs.csv

The last row is a run with nothing new: it loaded nothing and changed nothing, which is what makes a nightly schedule safe.

4. What was set aside, skipped or changed

Run	Source	Batch	What happened	Why
night2	menu	menu_items.xlsx	skipped	unchanged since the last load
night2	branches	branches.csv	skipped	unchanged since the last load
night2	ledger	gl_export.csv	skipped	unchanged since the last load
night2	ads	ads_export.csv	skipped	unchanged since the last load
night2	hr	database read	skipped	unchanged since the last load
night3	branches	branches.csv	skipped	unchanged since the last load
night3	pos	pos_B1_2026-09-30.csv	quarantined	required column missing: Net Sales
night3	pos	pos_B1_resent_copy.csv	skipped	same content as pos_B1_2026-09-20.csv
night3	pos	pos_B3_2026-09-30.csv	loaded with warnings	new column(s): Tip; stored in the raw layer
night3	ledger	gl_export.csv	skipped	unchanged since the last load
night3	ads	ads_export.csv	skipped	unchanged since the last load
night3	hr	database read	skipped	unchanged since the last load
rerun	menu	menu_items.xlsx	skipped	unchanged since the last load
rerun	branches	branches.csv	skipped	unchanged since the last load
rerun	pos	pos_B1_2026-09-30.csv	quarantined	required column missing: Net Sales; column(s) no longer sent: Tip
rerun	ledger	gl_export.csv	skipped	unchanged since the last load
rerun	ads	ads_export.csv	skipped	unchanged since the last load
rerun	hr	database read	skipped	unchanged since the last load

Source: tables/loads.csv

5. Is it right?

Every published table is tested after every build. The totals are carried from staging through every table to the cent.

Table	Test	Status	Detail
fact_sales	sales agree with staging	PASS	staging 1,485,821.00, fact 1,485,821.00
fact_daily_sales	daily totals agree with the lines	PASS	lines 1,485,821.00, daily 1,485,821.00
mart_kpi_monthly	monthly totals agree with the days	PASS	daily 1,485,821.00, monthly 1,485,821.00
stg_transactions	data no older than 2 days	PASS	latest 2026-09-30, 0 days old
stg_transactions	every site has the latest day	WARN	B1 last 2026-09-29

Source: tables/tests_last_night.csv

Table	PASS	WARN
dim_customer	2	0
dim_item	2	0
dim_place	2	0
fact_daily_sales	6	0
fact_expenses_monthly	1	0
fact_marketing_daily	1	0
fact_sales	6	0
fact_staff_monthly	1	0
mart_kpi_monthly	4	0
stg_transactions	1	1

Source: tables/tests_last_night.csv

The same models were then built on a second database (SQLite) from the same staging tables:

Table	On SQLite	Rows
dim_place	built	3
dim_item	built	9
fact_sales	built	39,723
dim_customer	built	899
fact_daily_sales	built	548
fact_expenses_monthly	built	78
fact_staff_monthly	built	18
fact_marketing_daily	built	549
mart_kpi_monthly	built	18

Source: tables/portability_models.csv

6. What the business gets

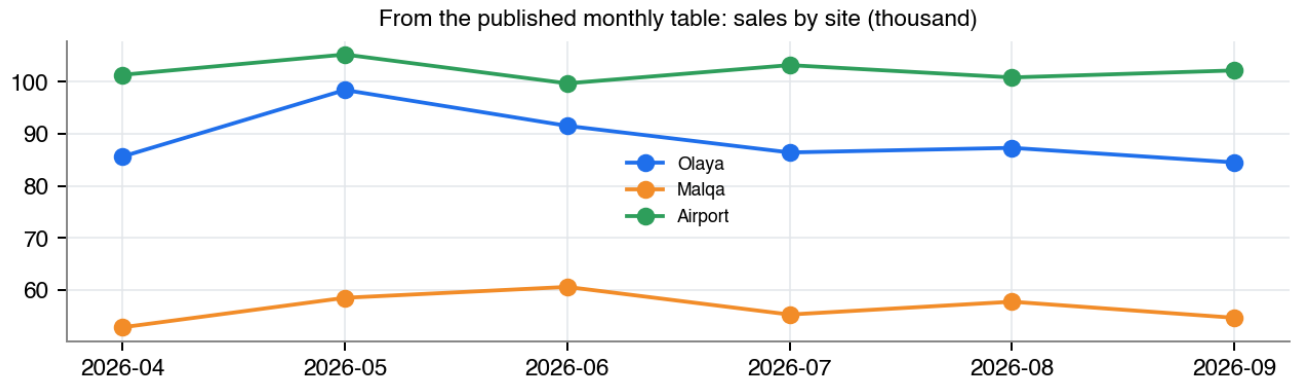


Figure 3. Read straight from a published table. Source: tables/mart_kpi_monthly.csv

The published tables, refreshed every night; what each one is for is in the connection guide and the data dictionary.

Table	Rows	Columns
dim_customer	899	5
dim_date	183	6
dim_item	9	6
dim_place	3	5
fact_daily_sales	548	9
fact_expenses_monthly	78	4
fact_marketing_daily	549	8
fact_sales	39,723	13
fact_staff_monthly	18	5
mart_kpi_monthly	18	10

Source: tables/layers.csv

7. Running it without us

The platform package holds the engine, the configuration, the models, a self-test on made-up data, schedules for Windows, macOS and Linux, and create-table scripts for PostgreSQL and BigQuery. It was unzipped into an empty folder and run with no access to the build environment: it passed.

Check	Status
cron entry has five time fields and a command	PASS
launchd plist parses	PASS
workflow parses and has a schedule	PASS
systemd timer has OnCalendar	PASS
Windows task command is complete	PASS

Source: tables/scheduler_checks.csv

8. All checks

Area	Test	Result	Threshold	Status
Loading	Status of the last night's run	warnings	ok passes; warnings warn; failed fails	WARN
Sources	Sources read directly (API, database, shared sheet)	17% (1 of 6)	pass >= 0.5; fail < 0.0	WARN
Loading	Batches set aside on the last night	11% (1 of 9)	pass <= 0.0; fail > 0.25	WARN
Correctness	Failed tests on the published tables	0	pass <= 0; fail > 0	PASS
Loading	Tables or sites behind the latest day	1 (B1 last 2026-09-29)	pass <= 0; fail > 99	WARN
Sources	Changes of format absorbed	1 (pos: Tip)	pass <= 0; fail > 99	WARN
Correctness	Batches loaded again when nothing new arrived	0	pass <= 0; fail > 0	PASS
Operations	Published columns with a written meaning	100%	pass >= 0.9; fail < 0.7	PASS
Operations	Seconds for a nightly run	1.0	pass <= 60; fail > 300	PASS
Correctness	Totals agree from staging to every published table	yes	must be yes	PASS
Operations	Same models built and tested on a second database (SQLite)	yes	must be yes	PASS
Operations	Platform package passes its self-test on a clean folder	yes	must be yes	PASS
Operations	Schedules for every platform check out	yes	must be yes	PASS
Operations	Create-table scripts check out	yes	must be yes	PASS
Security	Secrets written in the configuration	0	must be 0	PASS

Source: tables/tests.csv

Limitations and assumptions

- The platform has run on the demonstration nights and its own self-test. It has not run on the client's own servers or cloud; the first nights there should be watched.
- In this run the PostgreSQL script was checked on DuckDB and SQLite, and the BigQuery script for its form. The platform itself is tested by ProQuant on PostgreSQL and BigQuery; on the client's own database the first nights should still be watched.
- Updates are nightly or hourly, not live.
- A system's API is configured once, at its first client; a change of API by the provider needs a change of configuration.
- The platform checks that the data arrives whole and in its agreed form. Whether the source system recorded the business correctly is outside what it can see.

Appendix

Contracts on the first data received

Source	Batch	Status	Issues
pos	pos_B3_2026-09-26.csv	PASS	-
pos	pos_B3_2026-09-27.csv	PASS	-
pos	pos_B3_2026-09-28.csv	PASS	-
menu	menu_items.xlsx	PASS	-
branches	branches.csv	PASS	-
ledger	gl_export.csv	PASS	-
ads	ads_export.csv	PASS	-
hr	database read	PASS	-

Source: tables/contract_preview.csv

Mapping against the automatic suggestion

Source	Table	Field	Mapped to	Suggested	Agrees
pos	transactions	txn_id	Order No	Order No	True
pos	transactions	line_no	Line	Line	True
pos	transactions	date	Order Time	Order Time	True
pos	transactions	place_id	Branch	Branch	True
pos	transactions	channel	Channel	Channel	True
pos	transactions	item_id	Item Code	Item Code	True
pos	transactions	quantity	Qty	Qty	True
pos	transactions	amount	Net Sales	Net Sales	True
pos	transactions	party_id	Customer ID	Customer ID	True
pos	transactions	status	Status	Status	True
menu	items	item_id	Code	Item	False
menu	items	name	Item		False
menu	items	category	Category	Category	True
menu	items	unit_price	Price	Price	True
menu	items	unit_cost	Unit Cost	Unit Cost	True
branches	places	place_id	Branch	Branch	True
branches	places	name	Name	Name	True
branches	places	region	City	City	True
branches	places	opened	Opened	Opened	True
branches	places	capacity	Seats	Seats	True
branches	places	area_sqm	Area m2	Area m2	True
ledger	expenses	date	Posting Date	Posting Date	True
ledger	expenses	category	Account	Account	True
ledger	expenses	place_id	Cost Centre	Cost Centre	True
ledger	expenses	amount	Debit	Debit	True
ads	marketing	date	Day	Day	True
ads	marketing	channel	Platform	Platform	True
ads	marketing	campaign	Campaign	Campaign	True
ads	marketing	spend	Amount Spent	Amount Spent	True
ads	marketing	impressions	Impressions	Impressions	True
ads	marketing	clicks	Link Clicks	Link Clicks	True
ads	marketing	leads	Results	Results	True
hr	staff	date	month	month	True
hr	staff	place_id	branch	branch	True
hr	staff	headcount	employees	employees	True
hr	staff	hours	hours	hours	True
hr	staff	cost	total_pay	total_pay	True

Source: tables/mapping_review.csv

Data quality

Audit trail

Run ID	de_20261111T180225Z_effebd
Run started / finished (UTC)	2026-11-11T18:02:25+00:00
Analyst	info@proquantanalytics.com
Code version (git)	2703b230a84f (uncommitted changes) · core 0.1.0
Config SHA-256	b2812e6274f29bbb
Input: pos	90 rows · sha256 eeb8fb84b31112f2
Input: menu	9 rows · sha256 8800396b9d7ff08d
Input: branches	3 rows · sha256 58c51bee68e532a4
Input: ledger	78 rows · sha256 00c71258d3d80039
Input: ads	549 rows · sha256 378905229337afba
Input: hr	18 rows · sha256 16e3ff125107d878
Environment	python 3.13.5 · pandas 2.2.3 · numpy 2.1.3

Source: audit.json in the run folder

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